INCH-POUND

MIL-H-48358 (AR)
AMENDMENT 4
09 November 1993
SUPERSEDING
AMENDMENT 3
26 August 1992

MILITARY SPECIFICATION

HMX/RESIN EXPLOSIVE COMPOSITION LX-14-0 (For Use in Ammunition)

This Amendment forms a part of Military Specification MIL-H-48358 (AR), dated 30 September 1977, and is approved for use by the US Army Armament, Munitions and Chemical Command, and is available for use by all Departments and Agencies of the Department of Defense.

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Table I - after MIL-P-63196 add "Type I"

Para 3.1.1 Table I - after MIL-P-63196 add "Type I or III".

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3.2.3 Granulation. Correct "compsition" to "composition": and "specified-in" to "specified in".

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3.4 Foreign matter. Correct "woods" to "wood".

3.5 Add the following new paragraph:

"3.5 Vacuum stability test for colorizers. Any new or alternate colorizer(s) used to color the LX-14 lots (see 6.6) shall be subjected to a vacuum stability test (reactivity). The amount of gas evolved from the mixture of colorizer and the explosive shall be less than 3 ml when tested in accordance with 4.6.6."

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* 3.6 Add the following new paragraph:

"3.6 DTA/DSC explosive compatibility test for colorizer(s). Any new or alternate colorizer(s) used to color the LX-14 lots (see 6.6) shall be subjected to a Differential Thermal Analysis. If the resulting time versus delta T curve for the sample demonstrates no significant changes in the HMX transition and decomposition temperature when tested in accordance with 4.6.7, the colorizer passes this test."

4.3.1 Submission. Add "The sample shall be obtained from a production lot which has been produced by the contractor using the same production processes, procedures, and equipment as will be used in fulfilling the contract. All materials shall be obtained from the same sources of supply as will be used during regular production."

4.3.3 Rejection. Add "See MIL-A-48078".

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Para 4.4.1.b - Delete in its entirety and substitute:

"4.4.1.b - Binder from one manufacturer only. Not more than two blended acceptable binder lots of the same type produced in accordance with the same specification or specification revision under one continuous set of operating conditions. Mixing of Types I and III are not permitted."

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4.5 Testing. Add "PRECAUTION - This specification covers sampling and testing of toxic and hazardous materials. Accordingly, it is emphasized that all applicable safety rules, regulations, and procedures must be followed in the handling and processing of these materials."

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4.6.3.1 HMX Content. Delete 4.6.3.1.1 and 4.6.3.1.2 in their entirety and substitute the following:

"4.6.3.1.1 Reagent. Prepare the extraction solvent by adding excess RDX and HMX to 1, 2-dichloroethane (EDC). The mixture shall be maintained for at least eight hours at the temperature (± 2 degrees C) at which the analysis will be performed. Prior to use for extraction, the EDC solution shall be agitated for two hours and filtered.

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4.6.3.1.2 Procedure. Approximately 2.5g of the dried sample shall be accurately weighed (recorded to the nearest 0.1 mg) and quantitatively transferred to a 250 ml screw cap bottle. Add 150 ml of saturated EDC solution (see 4.6.3.1.1), attach the screw cap, and place the bottle on a mechanical shaker. The bottle shall be shaken at least thirty minutes, until visual examination indicates complete dissolution of the elastomeric binder. The sample solution shall then be transferred to a tared filtering crucible attached to a vacuum flask. About 30 ml of the saturated EDC solution shall then be poured into the bottle, the screw cap replaced, the bottle shaken by hand, and the contents transferred to the filtering crucible. This procedure shall be repeated at least twice in order to remove all residue from the bottle. The bottle and cap shall then be rinsed twice with 30 ml aliquots of the EDC solution, with the washings being transferred to the filtering crucible. The crucible shall then be removed from the flask, 20 ml of the saturated EDC solution shall be added, and the crucible shall be reconnected to the flask, allowing the liquid to filter through the residue. This procedure shall be performed a total of three times. The crucible shall then be removed from the flask and the interior portions beneath the glass frit shall be carefully rinsed with the saturated EDC. After rinsing, the crucible shall again be connected to the vacuum flask and aspirated for approximately two minutes. The crucible shall then be placed in an oven (100 ± 5 degrees C) for one hour, or until dry. After cooling to room temperature the crucible shall be accurately weighed (recorded to the nearest 0.1 mg). The HMX content of the sample shall be calculated as shown below:

$$\% \text{ HMX} = \frac{100\ (A-B)}{W}$$

A = Final weight of crucible, g

B = Tare weight of crucible, g

W = Sample weight, g"

* 4.6.6 Add the following new paragraph:

"4.6.6 Vacuum stability test (reactivity) for colorizer(s). The vacuum stability test for reactivity of the colorizer with a mixture of HMX and Estane in the proportions present in LX-14 shall be conducted at 100 degrees C for 40 hours in accordance with MIL-STD-650, method 504.1."

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* 4.6.7 Add the following new paragraph:

"4.6.7 DTA/DSC explosive compatibility test for colorizer(s). Differential Thermal Analysis shall be conducted by using a DTA or DSC apparatus, and shall be conducted on a sample of LX-14 manufactured with the proposed alternate colorizer. If a sample of LX-14 incorporating the proposed colorizer is not available, then a mixture of HMX, Estane, and the colorizer (in the proportions present in LX-14) may be used.

Place approximately 0.1 to 1.0 mg of the LX-14 sample into a DTA (or DSC) sample cup. Put this cup into the sample side of the DTA apparatus (or DSC apparatus). Put an empty cup into the reference side of the apparatus. Heat at a rate of 5 degrees C per minute.

Observe the resulting time vs delta T curve for any significant changes in HMX transition and decomposition temperature (i.e., extrapolated onset and peak temperature) compared to a time vs delta T curve obtained for HMX only using the same procedure and apparatus."

5. Delete section 5 in its entirety and replace with:

"5. PREPARATION FOR DELIVERY

5.1 PACKING

5.1.1 LEVEL A Packing shall be in accordance with Dwg. 7548644.

5.1.2 LEVEL B Packing shall be in accordance with Level A packing requirements.

5.1.3 LEVEL C Packing shall be in accordance with Dwg. 7548645 or Dwg. 9257923.

5.2 MARKING

5.2.1 LEVEL A Mark in accordance with Dwg. 7548644.

5.2.2 LEVEL B. Mark in accordance with Level A marking requirements.

5.2.3 LEVEL C Mark in accordance with Dwg. 7548645 or Dwg. 9257923."

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6.2: Delete all material following "DRDAR-QA, Dover, New Jersey".

6.2: Delete "DRDAR-QA, Dover, New Jersey" and substitute "SMCAR-QAR-R, Picatinny Arsenal, NJ 07806-5000".

6.3: Change "5703 F1" to "5702 F1 (MIL-P-63196 Type I" and delete "in accordance with MIL-P-63196".

6.3: Change "5702F1 (MIL-P-63196 Type I)" to: "...5702 (MIL-P-63196 Type I) or 5703P (MIL-P-63196 Type III)...".

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* 6.6 Colorizer. Delete paragraph 6.6 and substitute the following paragraph:

"6.6 Colorizer. The material shall be colorized with unadulterated finely powdered Pigment Violet #23/C.I. #51319 as identified in the Color Index Register Book published by the Society of Dryers and Colorists, Perkinhouse, West Yorkshire, England BD12JB, and issued in the U.S. by the American Association of Textile Chemists and Colorists, Research Triangle Park, NC 27709, and controlled chemically in the Chemical Abstracts as C.A.S. 6358-30-1. No extenders or additives to control color intensity shall be used, and the colorizer shall show less than one percent (1%) ash residue when heated to 600 +/-25 C for four (4) hours.

The colorizer shall be procured and certified to the aforementioned requirements for chemistry and purity. In addition, Vacuum Stability, and DTA/DSC explosive compatibility testing shall be performed on each lot of colorizer prior to use. These tests shall be documented and records retained in accordance with contract requirements.

Heliogen Violet Toner 49-6001 manufactured by General Aniline and Film Corp and Paliogen Violet Toner VT4414 manufactured by BASF Wyandotte have been found to be satisfactory when procured and certified to the aforementioned requirements for composition and purity. Prior to the use of any other colorizer, the results of the explosive compatibility tests shall be forwarded to U.S. Army Armament Research and Development Center, SMCAR-QAR-R and SMCAR-AEE-WE for review and approval."

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